

A Monte-Carlo simulation study of the depth-resolved backscattering photons in layered Human skin

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Abstract

The strong optical scattering of Human skin is still a great challenge in the field of non-invasive in-vivo imaging. To investigate the depth-resolved distribution of photons backscattered from deep Human skin tissues, we report a Monte-Carlo simulation model in a standard spatially offset configuration. In the simulation initializations, human skin was modelling to be composed of three layers (epidermis, papillary dermis, and reticular dermis), and each layer possessed inherent photon weights for backscattering intensity measurements. The simulation results clearly showed the total backscattered intensity varied with the increase of spatial offset, indicating depth-related photon information was encoded into spatial transients. We developed a nonlinear signal unmixture algorithm to recover the depth-resolved backscattered intensities through comparing the photon weights at different offsets. The proposed method paves the way for designing non-invasive depth-sensitive optical imaging and spectroscopic instrumentations for Human skin.

Monte-Carlo model of layered Human Skin

- Basic algorithm: Variance Reduction Monte-Carlo method.
- The parameters are shown in Tab.1. Flow chart is shown in Fig.2.
- Number of simulated photons: 10^6 .
- Results of different incidence angles are shown in Fig.1.
- Count the total backscattering photon weights associated with each layer (as shown in Figs.1 (b)-(c)).

Tab. 1 Parameters of Human Skin MC model

Parameters	Notation	Value
Refractive index of air	n_{air}	1.0
Refractive index of epidermis	n_{epi}	1.34
Refractive index of papillary dermis	n_{pap}	1.39
Refractive index of reticular dermis	n_{ret}	1.4
Refractive index of subcutaneous fat	n_{fat}	1.44
Wavelength(nm)	λ	800.0
Anisotropic factor	g	0.852
Absorption coefficient of epidermis(mm^{-1})	$\mu_{a,epi}$	0.2139
Absorption coefficient of papillary dermis(mm^{-1})	$\mu_{a,pap}$	0.0699
Absorption coefficient of reticular dermis(mm^{-1})	$\mu_{a,ret}$	0.0832
Reduced scattering coefficient of Human skin(mm^{-1})	μ_s'	22.973
Thickness of epidermis(mm)	d_{epi}	0.1
Thickness of papillary dermis(mm)	d_{pap}	0.2
Thickness of reticular dermis(mm)	d_{ret}	1.57

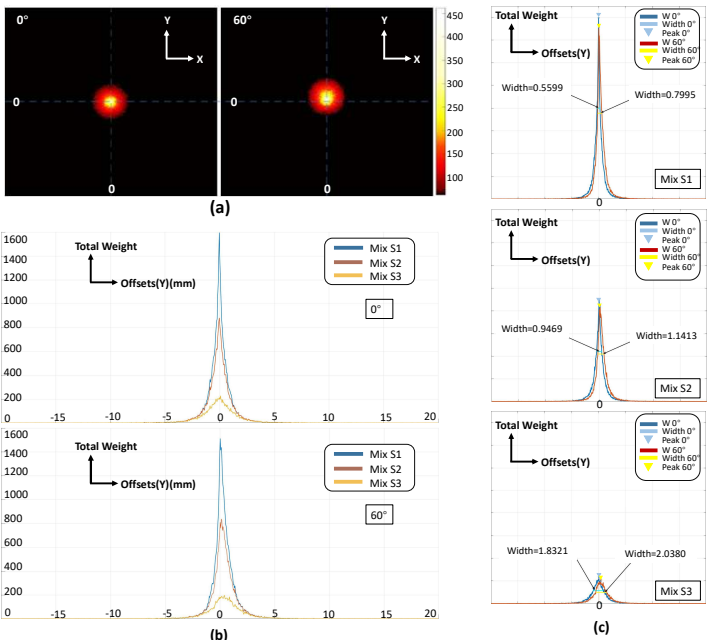


Figure 1. (a) Backscattering photon intensity profiles obtained at different incidence angles. (b) Total backscattering photon weights of each layer at different incidence angles, Mix S1, Mix S2, and Mix S3 indicate the layers of epidermis, papillary dermis and reticular dermis. (c) The comparisons of individual layers at different incidence angles.

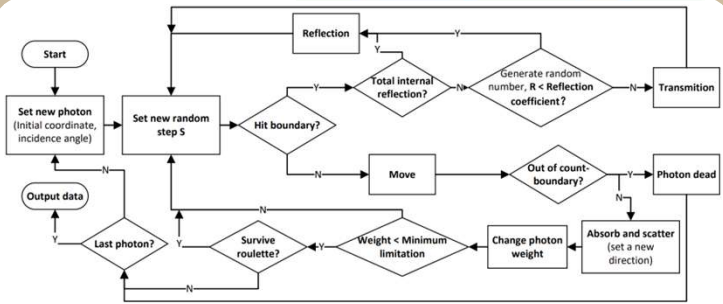


Figure 2. Flow chart of Monte Carlo simulation.

Unmixture algorithm and application

- Unmix the photon weight curve with ICA (Independent Component Analysis).
- Utilize the physical constrains and the results of ICA to make Nonlinear Programming.
- Multiple iteration, take the average, and smooth the curves.

Results of test and evaluation

- Generate test signals.
- Unmix the mixture of test signals by NICP (Nonlinear Independent Component Programming).
- Utilize correlation coefficient and integral to evaluate the accuracy of the unmixed signals (as shown in Fig.3).

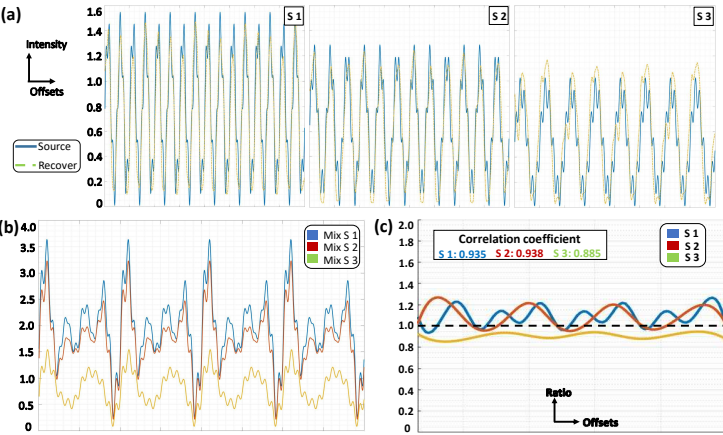


Figure 3. (a) Source and recovery signals. (b) Mixed signals for test. (c) Evaluation of the recovery signals, which is performed by the correlation coefficients and the integral ratios.

Application of the unmixture algorithm for depth resolving

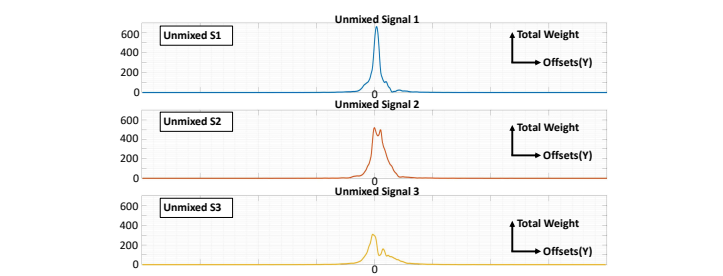


Figure 4. Unmixed depth-dependent photon weight signals. Signals 1-3 are from epidermis, papillary dermis and reticular dermis, respectively.

Conclusion

We proved the connection between backscattering photons and spatial offsets, and developed an unmixture algorithm for depth-resolved analysis of backscattering photons. The proportion matrix P of different layers could be used to unmix the scattering signals in real applications.

References

1. Lihong Wang et al. Computer Methods and Programs in Biomedicine, Volume 47, Issue 2, 1995, Pages 131-146.
2. Sun Y et al. Signal Processing. 2011;91(8):1838-51.
3. Saruwatari H et al. APSIPA Transactions on Signal and Information Processing. 2019;8(1).
4. Zhu C, Liu Q. Review of Monte Carlo modeling of light transport in tissues. J Biomed Opt. 2013;18(5):50902.
5. Salomatina E et al. J Biomed Opt. 2006 Nov-Dec;11(6):064026.
6. Leino AA et al. OSA Continuum. 2019;2(3).
7. Monakhova YB et al. Talanta. 2020;208:120451.

Acknowledgements

This work was supported by Shenzhen overseas high-level talent startup funding (No. JB11409014), Guangdong state funding (No. 2021A1515110030), and NSFC funding (No. 62205089).